Does worker mobility mediate knowledge spillovers?

León Tagle
September 22, 2026

Introduction

Firm spillovers, particularly as **knowledge diffusion**, are an important object in studies on:

- Firm competition (Bloom, Shankerman & Van Reenen, 2013).
- Growth (Akcigit & Ates, 2023).
- International trade (Alvarez, Buera, & Lucas, 2013).

Commonly conceptualized as occurring through **encounters** between individuals (Keller, 1998; Sampson, 2016; *the above*), rarely empirically micro-founded.

This paper: study and *horserace* two channels for knowledge diffusion between US R&D workers:

- **Co-location** (aka agglomeration gains).
- **Employee mobility** (aka poaching gains).

An *a fortiori* setting: if relevant anywhere, then here.

Data: advantages over previous studies



Data: advantages over previous studies

Inventors in patent data

- Long panel, reliable, linked to many other datasets.
- Focus on few individuals, best suited for *invention* rather than *diffusion*.
- Mobility data sparser and timing imprecise.



Data: advantages over previous studies



Longitudinal worker surveys

- Small representative sample.
- Not the right level of granularity:
  Either does not provide occupation (LEHD), or is aggregated to the industry level (CPS, SIPP, NLSY79/97, etc.).

Data: advantages over previous studies

This paper's data: Revelio Labs

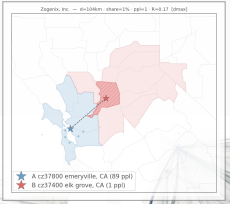
- Position-level data scraped from public LinkedIn profiles.
- Covariates include firm, occupation, location, and more.
- Result: 2007–2020 panel of **2.5M R&D workers**, **3.1M positions** at **3,335 listed, R&D-active firms**.



- Positions are linked to **publicly listed firms** through Revelio's own crosswalk to Compustat.
- **Occupations** are determined through Revelio-imputed O*NET codes + raw title token parsing.

Establishment-level data: from patents vs. Revelio

We use position locations to derive firm **establishments** (firm $\times$ commuting zone).





29,247 patent establishments and 90,125 inventor moves between them, 2007-2020



83,939 establishments and 465,583 scientist moves between them, 2007-2020

Spillovers: employee interactions mediated by firm "distance"

Reformulation of Bloom, Schankerman & Van Reenen (2013): sum over the spillovers firm i receives from every other firm:

$$SPILL_{i,t} = \sum_{j \neq i} DIST_{ij} \; n_{ij,t}$$

$DIST_{ij}$: normalized cosine similarity in:

- **Technology space**: patent subclass shares (TECH).
- **Product space**: sales shares by SIC segment (SIC).

$n_{ij,t}$: the interaction of firm j's workers with firm i

- **Co-location**: j's workers in the commuting zones of i's establishments.
- **Mobility**: workers who recently moved from j to i.

IV: oh, the disappointment

Mobility

✗ **Non-compete enforceability index**:
The big decisions fall outside our timeframe.

✗ **State R&D tax credit**:
Seemingly not enough cross-state variation — relevance.

✗ **Bartik-style dyadic instruments**:
- Without origin-firm FEs: OLS with extra steps;
- *With* origin-firm FEs: irrelevant, cannot predict single events.

Co-location

? **Moretti (2021)**: pending.

Still, here are some regressions

	(1)	(2)	(3)	(4)	(5)	(6)
	Mobility stock		Co-location		Both	
	TECH	TECH+SIC	TECH	TECH+SIC	TECH	TECH+SIC
Mobility stock d15, TECH						
Mobility stock d15, SIC						
Co-location, TECH						
Co-location, SIC						
N	16,875 / 16,259 / 16,312					

Within firm (firm and year FE), 2007–2020, SEs clustered by firm.
Cells, left to right: market value $\ln(V/A)$, productivity $\ln(sale/emp)$, patents (PPML).
positive, $p < 0.10$ positive, n.s. negative, $p < 0.10$ gap: negative, n.s.
Controls: own scientist mass ($\log(1+x)$), the turnover share, their interaction and R&D per scientist (asinh), lagged one year.
N: firm-years over 2,117 firms / 68,461 establishments (market value), 2,045 / 68,322 (productivity), 1,943 / 66,301 (patents); each regression keeps the firm-years where its outcome is observed.

Possible avenue: mobility interpreted as *gravity*

We could "instrument" (strong word) establishment-level mobility through a two-step estimator as in trade's gravity equation (Head & Mayer, 2014).

1. **Regress** establishment-*j*-to-*i* mobility m_{ji} on **origin and destination FEs** and **dyadic firm distance** (physical and abstract spaces) from 2007–2013:

$$\mathbb{E}[m_{ji,t}] = \exp\left(\alpha_{j,t} + \beta_{i,t} + \gamma_1 \ln dist_{ji} + \gamma_2 sameCZ_{ji} + \gamma_3 TECH_{ji} + \gamma_4 SIC_{ji}\right)$$

2. Project estimated origin and destination fixed-effects onto relevant covariates:

$$\hat{\beta}_{i,t} = \underbrace{\delta' Z_{i,t}}_{\substack{\text{R\&D/emp., turnover,} \\ \ln L,\ \text{turn.} \times \ln L}} + \underbrace{\theta_1 \bar{D}^{TECH}_{i,t} + \theta_2 \bar{D}^{SIC}_{i,t}}_{\substack{\text{mean proximity,} \\ \text{other side's active establishments}}} + \underbrace{\mu_{cz(i)}}_{\text{CZ FE}} + \mu_t + e_{i,t}, \qquad \text{likewise } \hat{\alpha}_{j,t}$$

3. Use **predicted dyadic mobility** from fitted baseline values as **instrument** for pooled 2014–2020 mobility.
4. Compute *TECH* and *SIC* spillovers out of it, aggregated to firm level, and run regressions on outcomes.

Results: gravity coefficients and instrumented pools

	Coefficient
Step 1: dyadic PPML, flow $j \to i$	
ln distance	-0.51^{***}
Same CZ	$+1.13^{***}$
TECH prox.	$+3.44^{***}$
SIC prox.	$+0.85^{***}$
Step 2: FE projections, destination / origin	
R&D/emp. ($100k)	-0.06^{***} / -0.04^{**}
Turnover	$+0.17$ / $+0.43^{**}$
ln scientists	$+0.45^{***}$ / $+0.56^{***}$
Turnover × ln sci.	$+0.24^{***}$ / -0.19^{***}
$\bar{D}$ TECH	-4.19^{***} / -4.56^{***}
$\bar{D}$ SIC	-2.41^{***} / -1.49^{***}
Pairs (step 1)	56.3M
N (step 2)	15,255 / 16,267
Within R^2	0.49 / 0.52

Left: step 1 is PPML with origin × year and destination × year effects, 2007–2013; step 2 has CZ and year effects. ***Right:*** 2014–2020 firm means (N 1,034 / 1,033), sic2 absorbed, robust SEs; (4) adds the actual co-location pools, uninstrumented; outcomes are market value $\ln(V/A)$, productivity $\ln(sale/emp)$, patents counts (run reduced form as PPML rather than 2SLS). * 90%, ** 95%, *** 99%; no SEs shown.

Instrumented	(1) TECH	(2) SIC	(3) Both	(4) Both + co-loc.
A. Market value, IV				
Mobility TECH	-0.02		-0.19^{***}	-0.30^{***}
Mobility SIC		$+0.06^{*}$	$+0.17^{***}$	$+0.23^{***}$
Co-loc. TECH				$+0.09^{***}$
Co-loc. SIC				-0.00
KP F	726	1,165	214	143
B. Productivity, IV				
Mobility TECH	$+0.02$		-0.17^{**}	-0.28^{***}
Mobility SIC		$+0.09^{**}$	$+0.19^{***}$	$+0.28^{***}$
Co-loc. TECH				$+0.09^{**}$
Co-loc. SIC				-0.03
KP F	724	1,165	214	144
C. Patents, reduced form				
Mobility TECH	$+0.17$		$+0.31$	$+0.47$
Mobility SIC		-0.01	-0.13	-0.21
Co-loc. TECH				-0.26
Co-loc. SIC				$+0.04$

Conclusion

Some conceptual points

- Interpreting results as **communicability** of firm knowledge.
- Two components of mobility: a **steady average** and **erratic realizations**.
- Difficult to separate **outflows** from **inflows** as permanent components.
- Co-location effects may require a **larger scale**.

Extensions and more

- **Leveraging patent data**: matching to non-listed firms; take to establishment-level.
- **Additional validation**: matching aggregates against NSF tallies; cross checking against matched inventors.
- **Employee-level distance metrics** from LinkedIn descriptions.
- **Heterogeneity** by industry + of the effect by distance bins as in Myers & Lanahan (2022).
- Maybe IVs can be saved…